

ReactJS HOL-1

1. Define SPA and its Benefits

A Single Page Application (SPA) is a web application that loads a single HTML page and dynamically updates its content without reloading the entire page. Instead of requesting a new page from the server for every action, only the required data is fetched.

Benefits of SPA:

- Faster page loading after the initial load.
 - Smooth and responsive user experience.
 - Reduced server requests.
 - Better performance because only required data is updated.
 - Easier integration with REST APIs.
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2. Define React and Identify its Working

React is an open-source JavaScript library developed by Facebook (Meta) for building interactive user interfaces, especially for single-page applications.

Working of React:

- React divides the user interface into reusable components.
 - Each component manages its own data (state).
 - When data changes, React updates only the affected part of the webpage.
 - React uses the Virtual DOM to make updates efficient and improve performance.
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3. Difference Between SPA and MPA

Single Page Application (SPA)	Multi Page Application (MPA)
Loads a single page.	Loads a new page for every request.

Single Page Application (SPA)	Multi Page Application (MPA)
Faster navigation.	Slower because pages reload.
Better user experience.	Traditional navigation.
Uses JavaScript frameworks like React, Angular, or Vue.	Mostly uses server-side rendering.
Suitable for modern web applications.	Suitable for large traditional websites.

4. Pros and Cons of Single Page Application

Advantages

- Fast navigation after the initial page load.
- Better user experience.
- Less bandwidth usage.
- Easy integration with APIs.
- Reusable UI components.

Disadvantages

- Initial loading time can be higher.
 - Search Engine Optimization (SEO) requires additional configuration.
 - JavaScript must be enabled.
 - Browser history management can be more complex.
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5. Explain About React

React is a component-based JavaScript library used for building modern web applications. It allows developers to create reusable UI components, making applications easier to develop and maintain.

Some important characteristics of React include:

- Component-based architecture.
 - High performance using Virtual DOM.
 - One-way data binding.
 - JSX syntax for writing HTML within JavaScript.
 - Strong community support and rich ecosystem.
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6. Define Virtual DOM

The Virtual DOM (Document Object Model) is a lightweight copy of the real DOM maintained by React.

When data changes:

1. React creates a new Virtual DOM.
2. It compares the new Virtual DOM with the previous one.
3. Only the changed elements are updated in the real DOM.

This process is called Diffing and Reconciliation, and it significantly improves application performance.

7. Features of React

Some important features of React are:

- **Component-Based Architecture:** Applications are built using reusable components.
- **Virtual DOM:** Improves performance by updating only modified elements.
- **JSX:** Makes UI development simple by combining HTML and JavaScript.
- **One-Way Data Binding:** Data flows from parent to child components, making applications predictable.
- **High Performance:** Efficient rendering through Virtual DOM.
- **Reusable Components:** Reduces code duplication.
- **Easy Learning Curve:** Simple syntax and excellent documentation.
- **Cross-Platform Development:** React Native enables mobile application development using React concepts.

Create a React application named "myfirstreact" and display the heading "Welcome to the first session of React".

Prerequisites

- Node.js
- npm
- Visual Studio Code

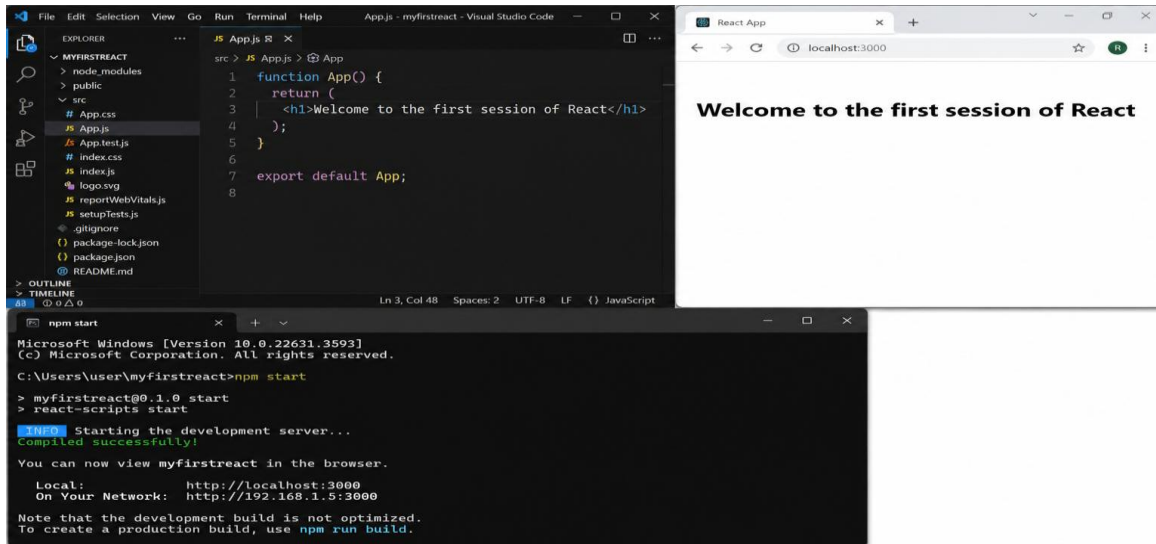
Implementation Steps

1. Install Node.js from <https://nodejs.org>.
2. Install Create React App: `npm install -g create-react-app`
3. Create the project: `npx create-react-app myfirstreact`
4. Open the project in Visual Studio Code.
5. Open `src/App.js` and replace its contents with the code below.
6. Run the application using `npm start`.
7. Open `http://localhost:3000` in a browser.

App.js

```
function App() {  
  return (  
    <h1>Welcome to the first session of React</h1>  
  );  
}  
  
export default App;
```

Output Screenshot:



Learning Outcome

Successfully created a basic React application, modified a React component, and executed the application using npm.

Conclusion

This exercise demonstrated the basic setup of a React application and verified the development environment by rendering a simple heading.